DeFi Student Protocol

- A Base Sepolia DeFi Super-App with AMM, ERC4626 vault, DAO governance, oracle checks, frontend, and subgraph.
- Built to demonstrate the full course stack in one auditable repository.

SCOPE

What We Built

- Upgradeable UUPS treasury controlled by Timelock.
- Factory with CREATE and CREATE2 AMM pair deployment.
- Constant-product AMM with LP token, 0.3% fee, and slippage protection.
- ERC20Votes + Permit, ERC4626 vault, ERC1155 items, Chainlink-compatible oracle.

Architecture

- React dApp talks to Base Sepolia through wagmi and viem.
- Subgraph indexes token, governance, AMM, vault, and item events.
- Governor schedules privileged calls through a 2 day Timelock.
- Timelock owns treasury roles, token owner, vault owner, factory, AMM pair, and oracle.

Smart Contracts

- GovernanceToken: ERC20Votes + ERC20Permit.
- YieldVault: ERC4626 deposit, mint, withdraw, redeem flows.
- ProtocolItems: ERC1155 with AccessControl and Pausable.
- AssemblyMath: Yul implementation benchmarked against Solidity.

AMM Design

- DefiSwapPair implements x*y=k from scratch.
- Swap input fee is 0.3%; a protocol fee share is sent to treasury.
- minAmountOut protects against slippage.
- Foundry invariant checks constant product behavior.

Governance Flow

- Voting delay: 1 day.
- Voting period: 1 week.
- Quorum: 4%. Proposal threshold: 1%.
- Lifecycle demonstrated: propose -> vote -> queue -> execute.

Security Posture

- Privileged functions use Ownable or AccessControl.
- Token and ETH movement uses SafeERC20, checked call, and ReentrancyGuard.
- No tx.origin authorization. No transfer/send for ETH.
- Vulnerability case studies reproduce and fix reentrancy and access-control bugs.

Oracle + Indexing

- Chainlink-compatible price adapter rejects stale, zero, negative, and incomplete rounds.
- MockV3Aggregator gives deterministic tests and demo deployment.
- Subgraph has more than 4 entities and documented GraphQL queries.
- Frontend has a dedicated subgraph data view.

Frontend

- Wallet connection through MetaMask injected connector.
- Reads balance, voting power, delegate address, AMM reserves, and vault state.
- Write actions: delegate, approve/deposit, cast vote.
- Readable errors for rejected transactions, wrong network, insufficient balance, and RPC issues.

Deployment

- All contracts deployed and verified on Base Sepolia.
- Post-deployment script verifies Timelock ownership, Governor settings, and treasury roles.
- Deployment addresses and BaseScan links are committed.
- Demo proposal is created and configured for the UI.

Testing + CI

- Hardhat tests pass locally.
- Foundry suite contains 80+ unit tests, 10 fuzz tests, 5 invariants, and 3 fork tests.
- CI runs compile, typecheck, Solidity lint, frontend Prettier, frontend build, subgraph build, tests, coverage, and Slither.
- Coverage gate is documented at 90%+ lines for contracts.

Defense Points

- Why Timelock is the final admin.
- How CREATE2 pair prediction works.
- Why ERC4626 rounding is delegated to OpenZeppelin and tested.
- How stale Chainlink rounds are rejected.
- What remains testnet-only: mock assets, mock feeds, centralized demo voting power.